TEAM XYZ REPORT

PHCs IN NIGERIA

YDP Datathon 2026

Outline

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1. INTRODUCTION

- 1.1 Background
- 1.2 Problem Statement
- 1.3 Purpose of the Project
- 1.4 Objectives

2. DATA AND METHODOLOGY

- 2.1 Data Sources
- 2.2 Data Pipeline Design
- 2.3 Data Model

4. EXPLORATORY DATA ANALYSIS

- 4.1 Overview
- 4.2 Operational Functionality and Risk Distribution
- 4.3 Staffing and Infrastructure Availability
- 4.4 Geographical Differences in Readiness and Risk
- 4.5 Medicine and Vaccine Shortages
- 4.6 Conclusion

5. CONCLUSION

6. RECOMMENDATIONS

1. Project Overview

- ## 1.1 Background

  Nigerian Primary Healthcare Centres (PHCs) serve as the first point of contact for many people seeking healthcare. According to a study report published in *BMJ Global Health* journal, PHCs account for approximately 85.3% of hospitals and clinics in the country, while more than 70% of Nigerians rely on primary healthcare facilities for basic healthcare services.

  These figures highlight the importance of understanding the current state of PHCs, including the gaps that may affect their ability to provide essential healthcare services.

- ## 1.2 Problem Statement

  It is important for health planners to have a reliable and current picture of what is happening within PHC facilities. Facility-level information on medicine and vaccine stocks, staffing, and service readiness should flow through Local Government Areas (LGAs) into the national health information system. However, reporting from facilities to the system can be inconsistent: some facilities report late, some incompletely, and some not at all.
  The result is a decision-making blind spot. Resource allocation, restocking, and staffing decisions may be based on outdated or incomplete information, allowing shortages at individual facilities to go unnoticed.

- ## 1.3 Purpose of the Project

  The project aims to develop a data-driven system that:

  1. Identifies the current operational status of each facility.
  2. Highlights PHCs with high risk of medicine, vaccine and staffing shortages.
  3. Supports data-driven decision-making for resource allocation, supply restocking and staffing interventions.

- ## 1.4 Objectives:

The objectives to be achieved are as follow:

1. Build an integrated PHC dataset covering facilities, staffing, medicines, vaccines and services.
2. Analyse patterns of PHC service readiness and resource availability.
3. Identify factors associated with medicine and vaccine storages and service disruption.
4. Develop a model that assigns a shortage-risk level to facilities.
5. Provide a dashboard for exploring high-risk facilities and geographic patterns.
6. Provide actionable information that can support health-planning decisions.

2. DATA AND METHODOLOGY

2.1 Data Sources

As of 2023, Nigeria had approximately 34,076 Primary Healthcare Centres (PHCs), of which an estimated 6,815 (20%) were considered fully functional. The source also highlights significant gaps in essential equipment and supplies required for PHC operations, including medicines and basic medical equipment. These figures provide the contextual baseline for this project.

The above information, obtained from *BMJ Global Health*, was used as a source-informed baseline for the design of the synthetic PHC dataset. The facility-level values generated for this project, including staffing levels, medicine and vaccine stocks, infrastructure, and service readiness, are synthetic and should not be interpreted as official facility-level statistics.

For the geographical structure of the dataset, Nigeria's administrative divisions were based on the *Nigeria Administrative Divisions dataset*, which identifies 37 administrative jurisdictions (36 states and the Federal Capital Territory), 774 Local Government Areas (LGAs), and 8,799 wards. The dataset also provides geographic coordinates and administrative relationships for these divisions.

These sources therefore provide the contextual and geographical foundation for the synthetic data used throughout the project, while the operational, staffing, inventory, and service-level observations are generated to support the analytical and predictive objectives of the study.

- ## 2.2 Data Pipeline Design

The pipeline follows a Medallion Architecture and consists of three components: Microsoft OneDrive (Ingestion Source), Microsoft Fabric Lakehouse (Storage), and Dataflow Gen2 / Notebook (Transformation Layer).

1. Ingestion Layer (OneDrive - Source)
A central repository is created on Microsoft OneDrive with a standardized folder structure: /PHC_Reports/{State}/{PHC_Name}/. Each PHC centre is granted write access only to its dedicated folder. PHCs upload their monthly/quarterly flat files (medicine, vaccine inventory, staffing) as CSV. A naming convention is enforced: PHCName_ReportType_YYYY-MM.csv to enable incremental processing.

2. Bronze Layer (Lakehouse Files - Raw)
A Fabric Lakehouse is provisioned. A OneDrive Shortcut is created within the Lakehouse Files section, pointing to the parent OneDrive repository. This provides a live, read-only pointer - when any PHC drops a file in OneDrive, it appears automatically in the Lakehouse Files/ directory without data duplication. This layer acts as immutable raw storage and audit trail.

3. Silver & Gold Layer (Lakehouse Tables - Curated)
Data is read from the Bronze Files via Dataflow Gen 2. The Dataflow performs:

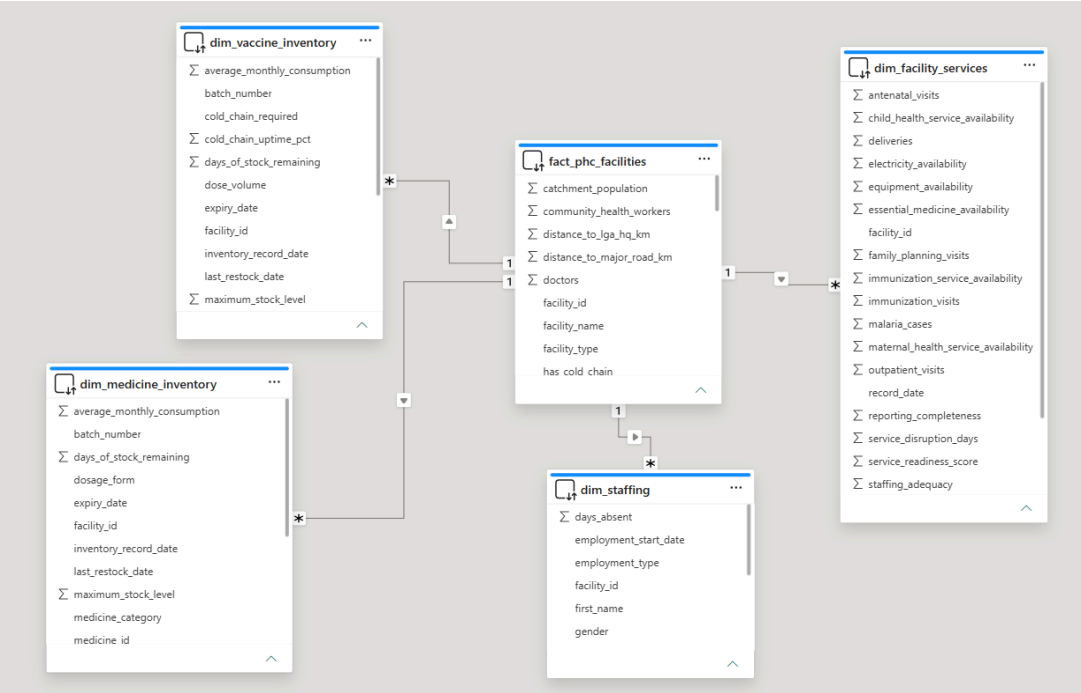
a) Schema enforcement & type casting

b)Normalization into curated tables: dim_vaccine_inventory, dim_staffing, fact_phc_facilities. These cleaned tables are loaded into the Lakehouse Warehouse (Silver/Gold layer).

4. Serving Layer (Semantic Model)
A Direct Lake semantic model is built on top of the Gold tables. Power BI dashboards consume this model directly, with scheduled refresh configured to pick up new files.

2.3 Data Model

The data is modelled using a star schema, with a central fact table containing PHC facility information and surrounding tables providing details on medicine inventory, vaccine inventory, staffing, and facility services. The model is designed to support efficient data analysis, reporting, and visualization in Power BI.



- **Fact Table:**
    - **fact_phc_facilities:** Stores core information about each PHC, including its location, ownership, operational status, infrastructure, catchment population, and healthcare worker counts. The table serves as the central reference for analysing facilities and their characteristics.
- **Dimension Tables:**
    - **dim_vaccine_inventory:** Contains facility-level vaccine inventory records, including stock levels, stock status, restocking dates, expiry dates, and cold-chain conditions. It supports analysis of vaccine availability and potential stockout risks.

- **dim_medicine_inventory:** Contains facility-level medicine inventory records, including medicine types, stock levels, minimum stock requirements, consumption, restocking dates, and expiry dates. It supports monitoring of medicine availability and shortages.
- **dim_staffing:** This table Stores individual staff records, including staff roles, qualifications, employment details, and other workforce attributes. It supports analysis of healthcare worker distribution and staffing adequacy across facilities.
- **dim_facility_services:** Contains monthly facility service records, including service utilisation, service availability, reporting completeness, disruptions, and service-readiness scores. It supports analysis of service delivery and changes in readiness over time.

4. EXPLORATORY DATA ANALYSIS

- ## 4.1 Overview

  This exploratory data analysis examines 34,076 PHC facility records to identify patterns in operational functionality, service readiness, staffing, infrastructure availability, and medicine and vaccine shortages. The objective is to establish where the most significant gaps occur, how risks differ across geographical areas, and which findings warrant further investigation.

  The analysis reveals three major concerns: a substantial proportion of facilities are classified as non-functional or partially functional, basic infrastructure is not consistently available, and vaccine shortages are more widespread than medicine shortages. These findings indicate that facility readiness should be assessed as a combination of operational capacity, workforce availability, infrastructure, and essential medical supplies.

  If the facility-level records are synthetic, the findings below describe the dataset rather than verified conditions across Nigeria's PHCs.

- ## 4.2. Operational Functionality and Risk Distribution

  Of the 34,076 facilities, 45% are classified as functional, 35% as non-functional, and 20% as partially functional. Consequently, 55% of facilities fall outside the fully functional category. This is the most prominent operational finding because it indicates that a majority of records do not represent facilities classified as fully functional.

  The distinction between non-functional and partially functional facilities is important. Non-functional facilities may

require investigation into whether essential services can be delivered at all, while partially functional facilities may be capable of delivering some services but have limitations. These groups should not automatically receive the same intervention without examining the reasons for their classification.

The risk distribution shows that 21.24% of facilities, or 7,237 facilities, are classified as high risk. Another 41.87%, representing 14,269 facilities, are medium risk, while 36.89% are low risk. The medium-risk group is the largest, indicating that potential vulnerability is not confined to a relatively small group of facilities.

This pattern suggests that risk management should address both immediate and emerging vulnerabilities. High-risk facilities warrant prompt assessment, while medium-risk facilities should be examined for deficiencies that could lead to further deterioration.

The highest displayed overall-risk values include PHC Gwagwalada at 10.3%, PHC Port-Harcourt at 8.9%, and PHC Kumbotso at 6.5%. For service risk, PHC Ikorodu records 9.06%, followed by PHC Gwagwalada at 7.89% and PHC Ede North at 6.59%. These results indicate that the facilities with the highest overall risk are not necessarily the same as those with the highest service risk. Different facilities may face different combinations of operational and resource-related constraints.

- ## 4.3 Staffing and Infrastructure Availability

  Staffing adequacy is 53.39%, while staffing risk is 46.61%. If staffing risk is calculated as the inverse of staffing adequacy, these values are mathematically consistent. The adequacy score nevertheless suggests that staffing requirements are not fully met under the measure's definition.

  The implications extend beyond the total number of workers available. A facility may have several healthcare workers but still lack the specific professional roles required to deliver essential services. The staffing gap should therefore be investigated by role and facility rather than assessed solely through an overall average.

  Infrastructure availability presents additional constraints. Water is reported at 49.68% of facilities, electricity at 50.07%, and storage-room availability at 58.18%. In other words, the reported availability of each of these resources is below 60%.

  Water and electricity are particularly notable because approximately half of the facilities are recorded as having these resources. Inadequate water may constrain sanitation and hygiene, while insufficient electricity may affect lighting, equipment operation, and cold-chain support. Storage-room availability is higher, but the presence of a room does not establish that medicines and vaccines are stored securely or under appropriate conditions.

  These findings also raise the possibility that facilities experience multiple constraints simultaneously. A facility lacking electricity may also have staffing gaps or shortages of essential supplies.

- ## 4.4 Geographical Differences in Readiness and Risk

  Service readiness varies across the states represented in the dataset. Lagos records the highest displayed readiness score at 75.60%, followed by Delta at 69.15%, Bayelsa at 67.96%, Ebonyi at 67.81%, and Edo at 67.56%.

  The difference between Lagos and Delta is 6.45 percentage points. This suggests that readiness is not uniform across the geographical areas represented. Lower readiness reflects differences in the underlying indicators.

  The selected state-level risk results provide further context. Lagos has an overall risk score of 21.33%, medicine risk of 14.60%, vaccine risk of 43.94%, readiness of 75.60%, and staffing adequacy of 91.76%. This is a comparatively favourable profile among the states shown.

  In contrast, Ondo records an overall risk score of 45.96%, medicine risk of 41.44%, vaccine risk of 61.82%, readiness of 64.21%, and staffing adequacy of 51.14%. Imo records an overall risk score of 43.89% and staffing adequacy of 44.66%. These results suggest different patterns of vulnerability: Ondo has comparatively high overall and medicine risk, while Imo has particularly low staffing adequacy among the selected states.

  Vaccine risk remains elevated across the states listed. For example, Enugu records 60.55%, Imo 60.67%, and Ondo 61.82%. This consistency suggests that vaccine availability may be a broader concern across the dataset rather than an issue restricted to a single state.

  Some states, including Oyo, Rivers, Osun, and Plateau, have blank medicine-risk values in the state matrix. These blanks reflect no records of medicine inventory.

- ## 4.5 Medicine and Vaccine Shortages

  Medicine shortages affect 76.61% of facilities, while vaccine shortages affect 90.45%. The corresponding shortage rates are 41.15% for medicines and 67.47% for vaccines. The vaccine shortage rate is therefore 26.32 percentage points higher than the medicine shortage rate.

  The facility percentages indicate how many facilities have at least one shortage, whereas the shortage rates describe the proportion of eligible inventory records classified as shortages under the calculation used. A facility with several shortages may contribute multiple records to a shortage-rate calculation but only once to a distinct-facility count.

  The results indicate that vaccine availability is the more prominent inventory concern in this dataset. The combination of a high vaccine shortage rate and a high proportion of facilities with at least one vaccine shortage suggests that the issue is both widespread across facilities and substantial among recorded vaccine entries.

  Finally, the inventory data represent a single reporting snapshot. The analysis identifies current patterns within that snapshot but cannot establish whether shortages are temporary, persistent, or worsening over time.

5. Conclusion

The analysis identifies a dataset characterised by substantial operational limitations, gaps in basic infrastructure, and widespread shortages of essential medical products. Although 45% of facilities are classified as functional, the remaining 55% are either partially functional or non-functional. Staffing adequacy stands at 53.39%, while water and electricity availability are each reported at approximately 50%.

The most prominent inventory finding is the higher prevalence of vaccine shortages compared with medicine shortages. Vaccine shortages affect 90.45% of facilities, compared with 76.61% for medicine shortages, and the vaccine shortage rate exceeds the medicine shortage rate by 26.32 percentage points. Geographical comparisons also show that risk and readiness vary across the states represented.

These findings suggest that vulnerability is multidimensional: facilities may face different combinations of staffing, infrastructure, service, and inventory constraints. Consequently, a single overall risk score is useful for identifying facilities for assessment, but the underlying risk components are necessary to determine what kind of intervention each facility may require.

6. Recommendations

For health authorities, PHC managers, and supply-chain decision-makers, the findings support the following actions:

1. Prioritise facilities for assessment. Begin with the 7,237 facilities classified as high risk. Review their individual staffing, infrastructure, readiness, medicine, and vaccine indicators before determining the appropriate response. Medium-risk

facilities should also be monitored to identify deteriorating conditions early.

2. Investigate vaccine supply gaps. Review stock availability for Meningitis A, HPV, Yellow Fever, IPV, and PCV, particularly in facilities reporting shortages. Establish whether the gaps are associated with replenishment delays, distribution issues, demand patterns, or incomplete inventory reporting before selecting interventions.

3. Address essential infrastructure and staffing needs. Verify facilities with no reported water or electricity and those lacking storage rooms. Assess staffing by professional role so that interventions target missing essential skills rather than relying only on total staff numbers.

4. Investigate state-level differences. Examine the factors associated with higher risk in states such as Ondo and Imo, and identify whether practices observed in better-performing areas can be adapted elsewhere. State averages should guide investigation rather than be treated as explanations in themselves.

5. Strengthen inventory reporting and monitoring. Record inventory consistently across facilities and reporting periods. Track product coverage, stockout rates, and the percentage of facilities with shortages separately so that decision-makers can distinguish missing information from confirmed stock availability.

6. Validate the risk model before allocating resources. Confirm that facility-level and aggregate risk scores use consistent scales, investigate blank medicine-risk values, and verify facility counts and denominator definitions. If the dataset is synthetic, validate the model using real facility records before using its rankings to make operational decisions.

7. Monitor whether interventions work. Once deficiencies have been verified and addressed, compare subsequent reporting periods to determine whether stock availability, staffing adequacy, infrastructure access, and service readiness improve.

These actions would allow stakeholders to move from identifying risk patterns to verifying facility needs, selecting targeted interventions, and monitoring whether those interventions lead to measurable improvements.